

Linear Algebra: Review

CS115 - Math for Computer Science

September 9, 2024

Roadmap

- Systems of Linear Equations
- Matrices
- Linear Independence
- Basis and Rank
- Linear Mappings

- For unknown variables $(x_1, \dots, x_n) \in \mathbb{R}^n$,

$$a_{11}x_1 + \dots + a_{1n}x_n = b_1$$

$$\vdots$$

$$a_{m1}x_1 + \dots + a_{mn}x_n = b_m$$

- Matrix representations:

$$\begin{pmatrix} a_{11} \\ \vdots \\ a_{m1} \end{pmatrix} x_1 + \dots + \begin{pmatrix} a_{1n} \\ \vdots \\ a_{mn} \end{pmatrix} x_n = \begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix} \iff \underbrace{\begin{pmatrix} a_{11} & \dots & a_{1n} \\ \vdots & & \vdots \\ a_{m1} & \dots & a_{mn} \end{pmatrix}}_A \underbrace{\begin{pmatrix} x_1 \\ \vdots \\ x_n \end{pmatrix}}_x = \underbrace{\begin{pmatrix} b_1 \\ \vdots \\ b_m \end{pmatrix}}_b$$

Identity Matrix and Matrix Properties

- A square matrix¹ I_n with $I_{ii} = 1$ and $I_{ij}=0$ for $i \neq j$, where n is the number of rows and columns. For example,

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_4 = \begin{pmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{pmatrix}$$

- Associativity:** For $A \in \mathbb{R}^{m \times n}$, $B \in \mathbb{R}^{n \times p}$, $C \in \mathbb{R}^{p \times q}$, $(AB)C = A(BC)$
- Distributivity:** For $A, B \in \mathbb{R}^{m \times n}$, and $C, D \in \mathbb{R}^{n \times p}$,
 (i) $(A + B)C = AC + BC$ and (ii) $A(C + D) = AC + AD$
- Multiplication with the identity matrix:** For $A \in \mathbb{R}^{m \times n}$, $I_m A = A I_n = A$

¹ # of rows = # of cols

Matrix: Addition and Multiplication

- For two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{m \times n}$,

$$\mathbf{A} + \mathbf{B} := \begin{pmatrix} a_{11} + b_{11} & \cdots & a_{1n} + b_{1n} \\ \vdots & & \vdots \\ a_{m1} + b_{m1} & \cdots & a_{mn} + b_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

- For two matrices $\mathbf{A} \in \mathbb{R}^{m \times n}$ and $\mathbf{B} \in \mathbb{R}^{n \times k}$, the elements c_{ij} of the product $\mathbf{C} = \mathbf{AB} \in \mathbb{R}^{m \times k}$ is:

$$c_{ij} = \sum_{l=1}^n a_{il}b_{lj}, \quad i = 1, \dots, m, \quad j = 1, \dots, k.$$

Inverse Matrix

- For a square matrix $\mathbf{A} \in \mathbb{R}^{n \times n}$, $\mathbf{B}$ is the **inverse** of $\mathbf{A}$, denoted by $\mathbf{A}^{-1}$, if

$$\mathbf{AB} = \mathbf{I}_n = \mathbf{BA}.$$

- Called **regular/invertible/nonsingular**, if it exists.
- If it exists, it is unique.

Transpose Matrix

- For a matrix $\mathbf{A} \in \mathbb{R}^{m \times n}$, $\mathbf{B} \in \mathbb{R}^{n \times m}$ with $b_{ij} = a_{ji}$ is the **transpose** of $\mathbf{A}$, which we denote by $\mathbf{A}^T$.

- Example.** For $\mathbf{A} = \begin{pmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}$,

$$\mathbf{A}^T = \begin{pmatrix} 0 & 1 & 0 \\ 2 & -1 & 1 \end{pmatrix}$$

- If $\mathbf{A} = \mathbf{A}^T$, $\mathbf{A}$ is called **symmetric**.

Inverse and Transpose: More Properties

- $AA^{-1} = I = A^{-1}A$
- $(AB)^{-1} = B^{-1}A^{-1}$
- $(A + B)^{-1} \neq A^{-1} + B^{-1}$
- $(A^T)^T = A$
- $(A + B)^T = A^T + B^T$
- $(AB)^T = B^T A^T$

Scalar Multiplication

- Multiplication by a scalar $\lambda \in \mathbb{R}$ to $\mathbf{A} \in \mathbb{R}^{m \times n}$

- **Example.** For $\mathbf{A} = \begin{pmatrix} 0 & 2 \\ 1 & -1 \\ 0 & 1 \end{pmatrix}$, $3 \times \mathbf{A} = \begin{pmatrix} 0 & 6 \\ 3 & -3 \\ 0 & 3 \end{pmatrix}$

- **Associativity**

- $(\lambda\psi)\mathbf{C} = \lambda(\psi\mathbf{C})$
- $\lambda(\mathbf{BC}) = (\lambda\mathbf{B})\mathbf{C} = \mathbf{B}(\lambda\mathbf{C}) = (\mathbf{BC})\lambda$
- $(\lambda\mathbf{C})^T = \mathbf{C}^T\lambda^T = \mathbf{C}^T\lambda = \lambda\mathbf{C}^T$

- **Distributivity**

- $(\lambda + \psi)\mathbf{C} = \lambda\mathbf{C} + \psi\mathbf{C}$
- $\lambda(\mathbf{B} + \mathbf{C}) = \lambda\mathbf{B} + \lambda\mathbf{C}$

Solving Systems of Linear Equations

- Start as usual by getting echelon form.

$$\begin{array}{rcl}
 x + y - z = 2 & & x + y - z = 2 \\
 2x - y = -1 & \xrightarrow{-2\rho_1 + \rho_2} & -3y + 2z = -5 \\
 x - 2y + 2z = -1 & \xrightarrow{-1\rho_1 + \rho_3} & -3y + 3z = -3
 \end{array}
 \quad
 \xrightarrow{-1\rho_2 + \rho_3}
 \quad
 \begin{array}{rcl}
 x + y - z = 2 \\
 -3y + 2z = -5 \\
 z = 2
 \end{array}$$

- Make all the leading entries one.

$$\begin{array}{rcl}
 x + y - z = 2 \\
 \xrightarrow{(-1/3)\rho_2} & & y - (2/3)z = 5/3 \\
 & & z = 2
 \end{array}$$

- Finish by using the leading entries to eliminate upwards, until we can read off the solution.

$$\begin{array}{rcl}
 x + y - z = 2 & & x + y = 4 \\
 y - (2/3)z = 5/3 & \xrightarrow{\rho_3 + \rho_1} & y = 3 \\
 z = 2 & \xrightarrow{(2/3)\rho_3 + \rho_2} & z = 2
 \end{array}
 \quad
 \xrightarrow{-\rho_2 + \rho_1}
 \quad
 \begin{array}{rcl}
 x & = & 1 \\
 y & = & 3 \\
 z & = & 2
 \end{array}$$

Algorithms for Solving System of Linear Equations

1. Pseudo-inverse

$$\mathbf{Ax} = \mathbf{b} \iff \mathbf{A}^T \mathbf{Ax} = \mathbf{A}^T \mathbf{b} \iff \mathbf{x} = (\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T \mathbf{b}$$

- $(\mathbf{A}^T \mathbf{A})^{-1} \mathbf{A}^T$: *Moore-Penrose pseudo-inverse*
- many computations: matrix product, inverse, etc

2. Gaussian elimination

- intuitive and constructive way
- cubic complexity (in terms of # of simultaneous equations)

3. Iterative methods

- practical ways to solve indirectly
- (a) stationary iterative methods: Richardson method, Jacobi method, Gaus-Seidel method, successive over-relaxation method
- (b) Krylov subspace methods: conjugate gradients, generalized minimal residual, biconjugate gradients

Linear Combinations of Linearly Independent Vectors

- Vector space V with k linearly independent vectors $\mathbf{b}_1, \mathbf{b}_2, \dots, \mathbf{b}_k$
- m linear combinations $\mathbf{x}_1, \mathbf{x}_2, \dots, \mathbf{x}_m$. (Q) Are they linearly independent?

$$\mathbf{x}_1 = \lambda_{11}\mathbf{b}_1 + \lambda_{21}\mathbf{b}_2 + \dots + \lambda_{k1}\mathbf{b}_k$$

$$\vdots$$

$$\mathbf{x}_m = \lambda_{1m}\mathbf{b}_1 + \lambda_{2m}\mathbf{b}_2 + \dots + \lambda_{km}\mathbf{b}_k$$

$$\mathbf{x}_j = \overbrace{(\mathbf{b}_1, \dots, \mathbf{b}_k)}^B \overbrace{\begin{pmatrix} \lambda_{1j} \\ \vdots \\ \lambda_{kj} \end{pmatrix}}^{\lambda_j}, \quad \mathbf{x}_j = B\lambda_j$$

- $\sum_{j=1}^m \psi_j \mathbf{x}_j = \sum_{j=1}^m \psi_j B\lambda_j = B \sum_{j=1}^m \psi_j \lambda_j$
- $\{\mathbf{x}\}$ linearly independent $\iff \{\lambda\}$ linearly independent

Generating Set and Basis

- **Definition.** A vector space $V = (\mathcal{V}, +, \cdot)$ and a set of vectors $\mathcal{A} = \{\mathbf{x}_1, \dots, \mathbf{x}_k\} \subset \mathcal{V}$.
 - If every $v \in \mathcal{V}$ can be expressed as a linear combination of $\mathbf{x}_1, \dots, \mathbf{x}_k$, $\mathcal{A}$ is called a **generating set** of V .
 - The set of all linear combinations of $\mathcal{A}$ is called the **span** of $\mathcal{A}$.
 - If $\mathcal{A}$ spans the vector space V , we use $V = \text{span}[\mathcal{A}]$ or $V = \text{span}[\mathbf{x}_1, \dots, \mathbf{x}_k]$
- **Definition.** The minimal generating set $\mathcal{B}$ of V is called **basis** of V . We call each element of $\mathcal{B}$ **basis vector**. The number of basis vectors is called **dimension** of V .
- **Properties**
 - $\mathcal{B}$ is a maximally² linearly independent set of vectors in V .
 - Every vector $x \in V$ is a linear combination of $\mathcal{B}$, which is unique.

²Adding any other vector to this set will make it linearly dependent.

Vector/Matrix Form

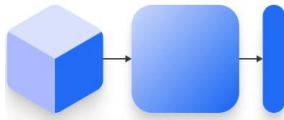
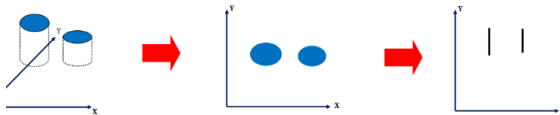
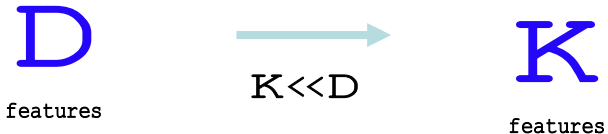
$$b + w_1x_1 + w_2x_2 + \cdots + w_Dx_D; b, w_i, x_i - \text{scalar}$$

$$\sum_{k=1}^K z_{nk} \mathbf{u}_k; z_{nk} - \text{scalar}; \mathbf{u}_k - \text{vector}$$

$$b + w_1x_1 + w_2x_2 + \cdots + w_Dx_D = b + \mathbf{w}^T \mathbf{x}$$

$$\sum_{k=1}^K z_{nk} \mathbf{u}_k = \mathbf{W} \mathbf{z}_n$$

Dimension Reduction



Dimension Reduction

$$\mathbf{x}_n \approx \sum_{k=1}^K z_{nk} \mathbf{u}_k = \mathbf{W} \mathbf{z}_n$$

$\mathbf{W}$ – linear mapping high-dim $\mathbf{x}_n$
into low-dim $\mathbf{z}_n$

$\mathbf{W} = [\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_K]$ is $D \times K$

$\mathbf{z}_n = [z_{n1}, z_{n2}, \dots, z_{nK}]$ is $K \times 1$

Technique f such that $\mathbf{z}_n = f(\mathbf{x}_n)$

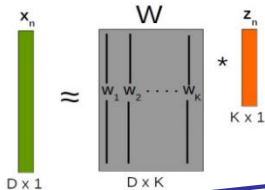


- $\mathbf{x}_n \in \mathbb{R}^D$, $n = 1, 2, \dots, N$ linear combination of K basic vectors $\mathbf{w}_1, \mathbf{w}_2, \dots, \mathbf{w}_K \in \mathbb{R}^D$ ($K \ll \min\{D, N\}$)
- Each $\mathbf{x}_n \in \mathbb{R}^D$ is replaced by K - $\mathbf{z}_n$ vector
- To present N vectors $\{\mathbf{x}_n\}_{n=1}^N$ require $\mathbf{W}$ and $\{\mathbf{z}_n\}_{n=1}^N$
 - Before Dimension Reduction: $N \times D$
 - After Dimension Reduction $N \times K + D \times K = (N + D) \times K$
($K \ll \min\{D, N\}$)

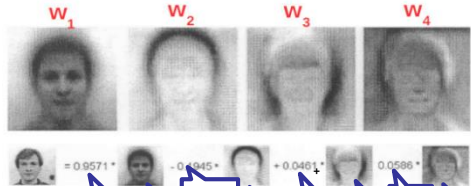
■ Face images

Mỗi ảnh cơ sở có thể xem là một khuôn mẫu chứa những đặc trưng/thuộc tính phổ biến của khuôn mặt

$K=4$ "basis" khuôn mặt cơ sở



01 image $x_n \in \mathbb{R}^D$



z_{n1}

w_1

z_{n2}

w_2

z_{n3}

w_3

z_{n4}

w_4

Dimension Reduction